Exp : 3 **Write the Python Program for Water Jug Problem**

Aim:

To write a python program for water jug problem.

Algorithm:

Step1: Start with two empty jugs of different capacities, and a target quantity of water to measure.

Step2: Fill one jug to its maximum capacity.

Step3: Pour the water from the full jug into the other jug until it is full or the first jug is empty.

Step4: If the second jug is full, pour out the water to empty it.

Step5: Repeat steps 2-4, filling and pouring water between the two jugs, until the desired quantity of water is measured or all possible combinations are tried.

Program:

```
from collections import deque
def water_jug(jug1, jug2, target):
    visited = set()
    queue = deque([((0, 0), [])])
    while queue:
        (a, b), path = queue.popleft()
        if (a, b) in visited:
            continue
        visited.add((a, b))
        path = path + [(a, b)]
        if a == target or b == target:
            print("Solution Path:")
            for state in path:
                print(state)
```

```
            return
        next_states = [
            (jug1, b),                # Fill Jug1
            (a, jug2),                # Fill Jug2
            (0, b),                   # Empty Jug1
            (a, 0),                   # Empty Jug2
            (max(0, a-(jug2-b)), min(jug2, a+b)),  # Jug1 -> Jug2
            (min(jug1, a+b), max(0, b-(jug1-a)))   # Jug2 -> Jug1
        ]
        for state in next_states:
            if state not in visited:
                queue.append((state, path))
    print("No Solution Exists.")
# Example
jug1 = 4
jug2 = 3
target = 2
water_jug(jug1, jug2, target)
```

Output:

```
>>>
Solution Path:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>
```

Result:

The output has verified.